INTRUSION DETECTION SYSTEM

USING CONVOLUTIONAL NEURAL NETWORKS

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MINI PROJECT | SEMESTER - 6

OVERVIEW

Motivation

Intrusion Detection is an important component of infrastructure protection mechanisms. Given the increasing complexities of today's network environments, more and more hosts are becoming vulnerable to attacks and hence it is important to look at systematic, efficient and automated approaches for Intrusion Detection.

OVERVIEW

Abstract

System administrators can benefit from deploying Network Intrusion Detection Systems (NIDS) to find potential security breaches. There are many challenges to develop a flexible and effective NIDS in order to prevent high false alarm rates and low detection accuracy against unknown attacks. In this paper, we propose a deep learning method to implement an effective and flexible NIDS. We propose the use of a convolutional neural network (CNN), an advanced deep learning technique.

Dataset Used – CICIDS2017

The data capturing period started at 9 a.m., Monday, July 3, 2017 and ended at 5 p.m. on Friday July 7, 2017, for a total of 5 days. Monday is the normal day and only includes the benign traffic. The implemented attacks include Brute Force FTP, Brute Force SSH, DoS, Heartbleed, Web Attack, Infiltration, Botnet and DDoS. They have been executed both morning and afternoon on Tuesday, Wednesday, Thursday and Friday.

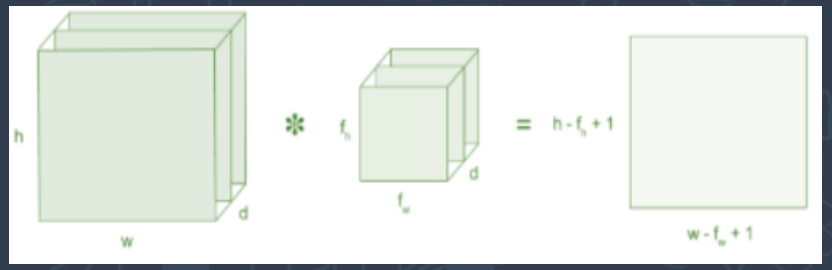
Dataset Description :

- Labelled Data
- Variety of Operating Systems
- All common available protocols such as HTTP, HTTPS, FTP, SSH & E-Mail Protocols
- 15 Most common attacks from 2016 McAfee Report
- 78 Network-flow features

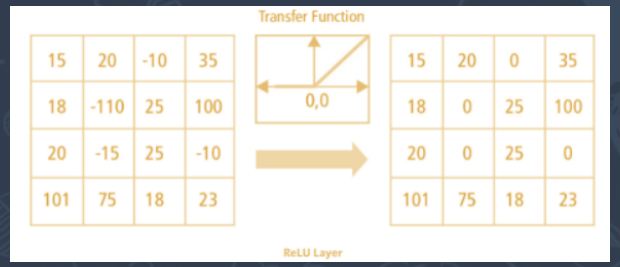
Attack Labels

Serial	Attack Label	Instances (Million)	Short Description
1.	BENIGN	2.359 M	Normal Traffic
2.	Bot	0.002 M	Utilizes a series of connected computers for attack
3.	DDoS	0.042 M	Multiple systems flood the bandwidth of the target system
4.	DoS GoldenEye	0.010 M	Used to perform security testing for research purposes
5.	DoS Hulk	0.231 M	Generate volumes of unique traffic, Bypassing the cache engine
6.	DoS Slow-httptest	0.055 M	Testing tool that simulates Application Layer
7.	DoS Slowloris	0.058 M	Generates traffic by sending partial requests, tries to hold connection for a long time
8.	FTP Patator	0.079 M	A type of brute force attack, generates large number of consecutive guesses for cracking FTP passwords
9.	Heartbleed	Neg.	A bug in the OpenSSL cryptographic software library, It allows stealing info encrypted under SSL/TLC
10.	Infiltration	Neg.	A piece of malicious software that tries to enter the system like viruses, worms, adwares
11.	Port Scan	0.159 M	By using a listening service to see what ports are open
12.	SSH Patator	0.006 M	A type of brute force attack, generates large number of consecutive guesses for cracking SSH passwords
13.	WA – Brute Force	0.002 M	A type of brute force attack, generates large number of consecutive guesses for cracking web passwords
14.	WA – SQL Injection	Neg.	By execution malicious SQL Statements
15.	WA – XSS	Neg.	A way of bypass the SOP (Same origin policy) Concept which forbids a web application to retrieve content from other origin

CNN Architecture



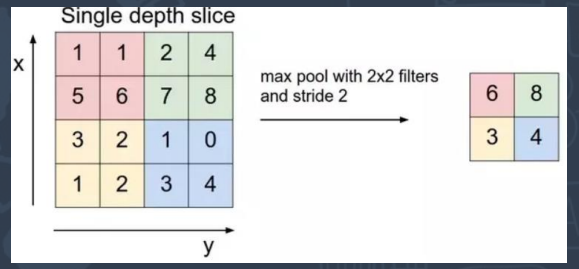
Convolution



15	20	-10	35
18	-110	25	100
20	-15	25	-10
101	75	18	23

15	20	0	35
18	0	25	100
20	0	25	0
101	75	18	23

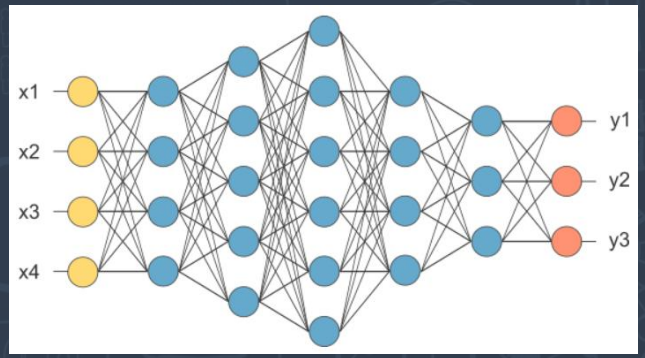
ReLU



1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

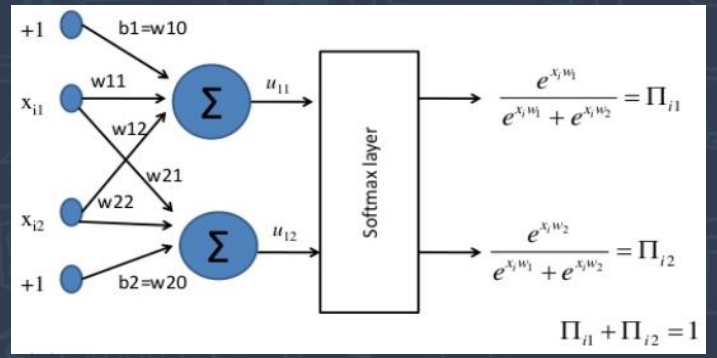
6	8
3	4

Max Pooling

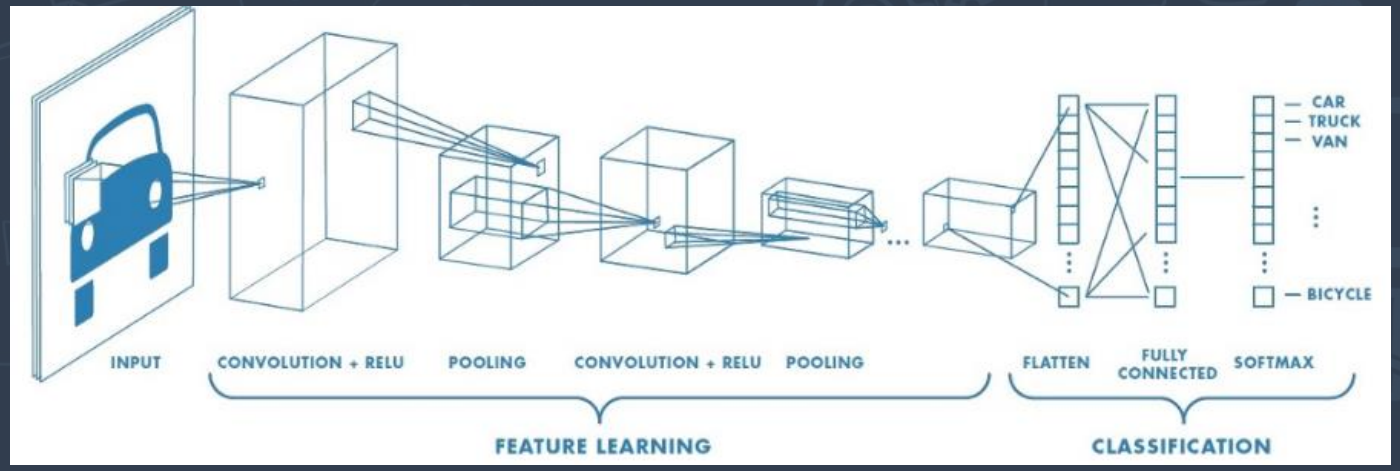


Fully Connected

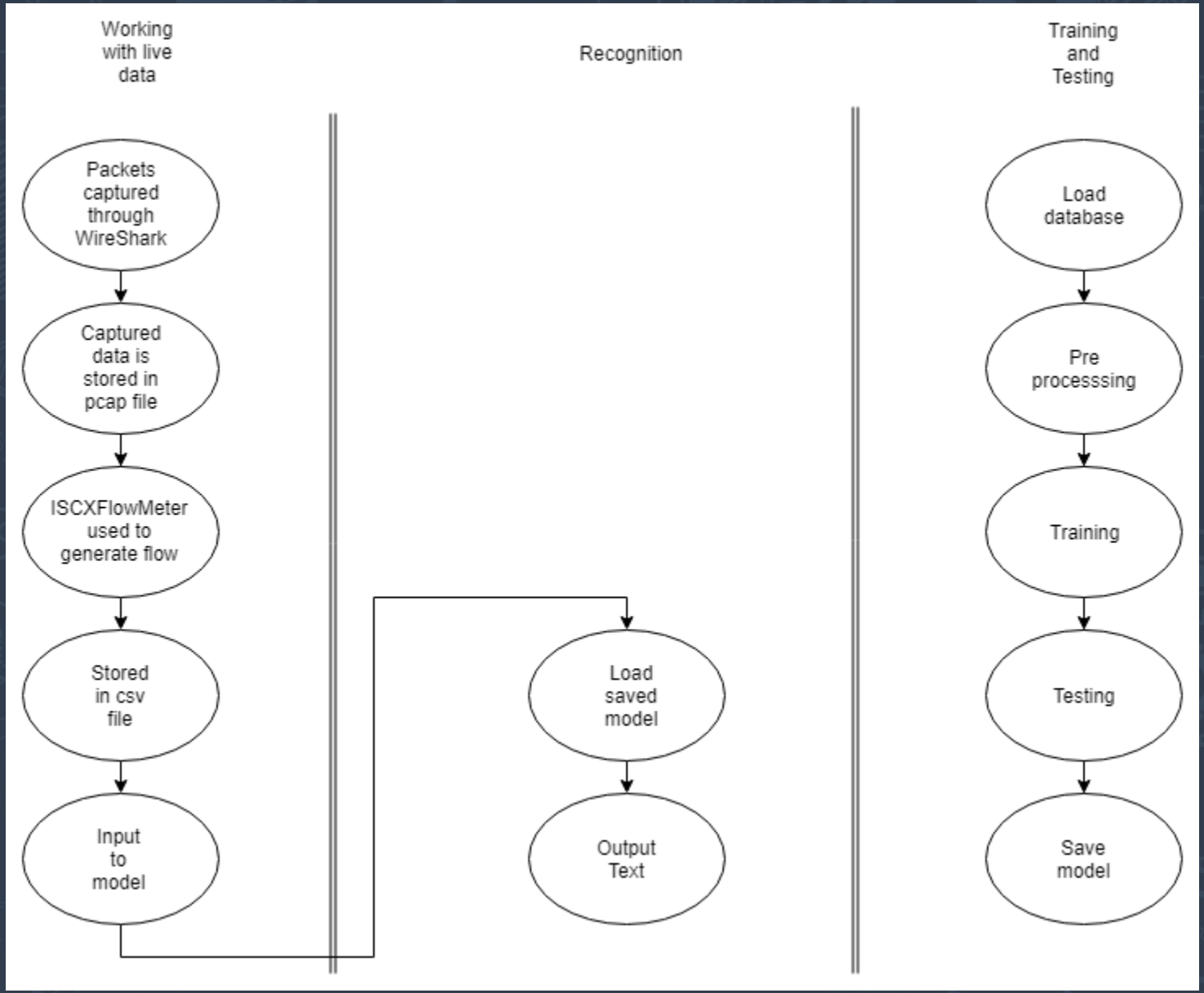
Softmax



Whole CNN Architecture



WORKFLOW


Working with live data
Recognition
Training and Testing
Packets captured through WireShark
Captured data is stored in pcap file
ISCXFlowMeter used to generate flow
Stored in csv file
Input to model
Load saved model
Output Text
Load database
Pre processsing
Training
Testing
Save model

METHODOLOGY

PRE-PROCESSING

CICIDS2017 consists of data packets with 78 network attributes. In order to make it a 9*9 image for input into our CNN, we appended 3 additional features and the resulting image was normalised. Hence, the input data would be an image of some fixed size.

LIVE DATA

Live data packets were captured using Wireshark through the Npcap Loopback adapter which successfully captures data flowing into server hosted in the system. Npcap is able to sniff loopback packets (transmissions between services on the same machine) by using the Windows Filtering Platform (WFP). The resultant captures data is stored in pcap files. These pcap files are used for generation of network flow data.

MODEL

We have an input layer with 20 convolutional filters and kernel size 2x2. The input is of volume (9,9,1). This layer will produce an output of volume (9,9,20). Then, we used a maxpool layer with pool size (2,2) which downsamples the feature map to (4,4,20). This layer is followed by another layer of 10 filters which constructs a feature map of shape (4,4,10). Then, we used a maxpool layer with pool size (2,2) which downsamples the feature map to (2,2,10). Next a dropout layer with probability 0.25 is added to reduce overfitting. Hence, our final feature map of size (4,4,20). ReLu activation is used in both layers. This feature map is then used to classify each image into their respective characters using fully connected layers. We have used 2 dense layers with 128 neurons in the first one and with 64 neurons in the second one. The first layer uses Relu activation whereas the second layer uses softmax activation.

CONCLUSION

End Product

The end product of this project is be a system which is used to detect intrusions in network flows. It is also used to detect live network data an detect an anomaly whenever one occurs. Also, the model gets updated on live data, according to the needs of the user. The overall accuracy of the model is about 95 percent. The true negative rate was about 0.97 and hence the false positive rate was about 0.03.

REFERENCES

References

CICIDS2017 - https://www.unb.ca/cic/datasets/ids-2017.html

Understanding CNN - https://towardsdatascience.com/applied-deep-learning-part-4-convolutional-neural-networks-584bc134c1e2

Research Papers -

- T.Mirza – Building an Intrusion Detection System
- Kovacs, Alzubi – Evaluation of machine learning algorithms for intrusion detection system
- S. Paliwal , R. Gupta – An Artificial Neural Network for network intrusion detection system